

RUNMAXXIN: Adaptive Music Recommendation for Runners

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Abstract

RUNMAXXIN is an edge-based system that adapts music to a runner’s intent and physical state. We propose a neuro-symbolic pipeline that combines SetFit for intent classification with real-time biometric analysis. A symbolic gate enforces safety rules, overriding tracks if the runner’s heart rate exceeds critical thresholds. Our results demonstrate that this hybrid approach outperforms heuristic baselines, ensuring performance-oriented music within safe physiological limits.

1 Introduction

Music acts as a natural pacer for athletes. However, existing apps ignore the gap between a runner’s mental goal (e.g., “easy recovery”) and their actual physiological effort (e.g., “high heart rate”). Current solutions are either opaque “black-box” models or simple heuristics that match tempo to heart rate without understanding the user’s intent.

We address this by building an AI pipeline that fuses Natural Language Understanding (NLU) with biometric sensor data. Our contribution includes a SetFit-based classifier for intent, a controller to fuse semantic and physical signals, and an OWL-based symbolic gate for safety. This report details the pipeline architecture, its performance against baselines, and our strategy for edge-compatible, real-time music recommendation.

2 Related Work

Early systems like MPTrain [1] and PAPA [2] pioneered physiological music steering but required rigid, numeric goals. Modern deep learning models [3] improve preference prediction but are too heavy for edge devices and lack interpretability. We adopt SetFit [4] for lightweight, few-shot intent classification: it reaches high accuracy through contrastive fine-tuning on few-shot pairs (Fig. 1) and stays competitive even in very low-data regimes (Fig. 2), avoiding the latency of large transformers such as RoBERTa-large [6]. Unlike COMUS [5], which uses ontologies as a primary recommender, we use an OWL-based symbolic gate strictly as a safety filter to provide hard guarantees.

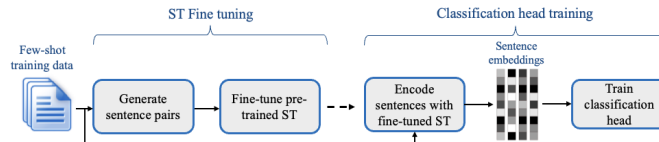


Figure 1: The SetFit training scheme: contrastive fine-tuning of a Sentence Transformer on few-shot sentence pairs.

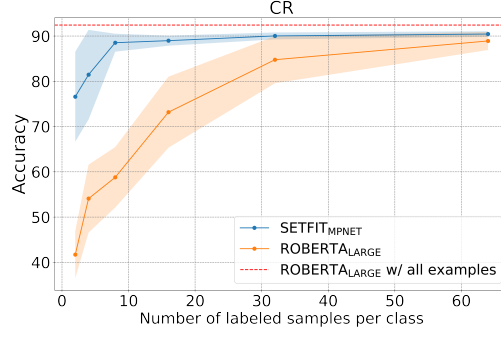


Figure 2: SetFit accuracy in the low-data regime compared to standard fine-tuning.

3 Methodology

The RUNMAXXIN pipeline follows a modular neuro-symbolic flow (Fig. 3).

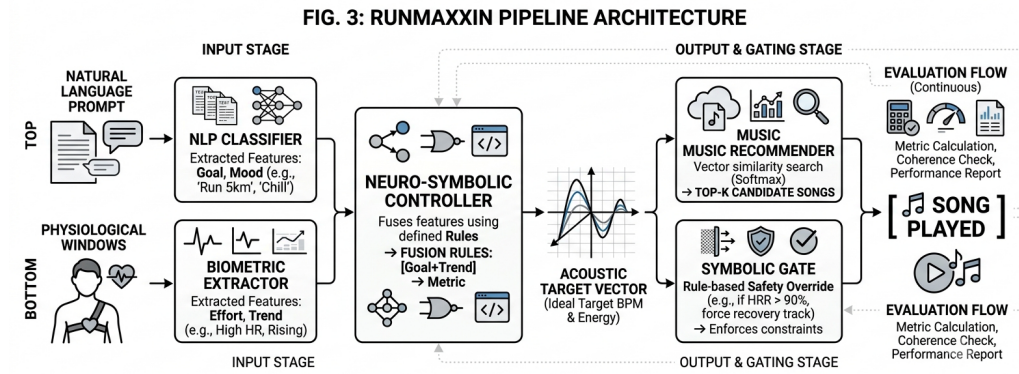


Fig. 3. Detailed RUNMAXXIN pipeline architecture, illustrating dual-path feature extraction (NLP and Biometric), neuro-symbolic fusion, recommendation, symbolic safety gating, and evaluation loop.

Figure 3: RUNMAXXIN pipeline: Intent extraction, sensor processing, and symbolic safety gating.

- **Intent Extraction:** A SetFit classifier parses user text to identify goals and moods (latency < 16ms).
- **Biometric Processing:** Sensor data are grouped into 30s windows to calculate heart rate and effort trends.
- **Neuro-Symbolic Controller:** Fuses these inputs into an acoustic target vector $[BPM, Energy, Valence]$.
- **Recommender:** Ranks tracks by Euclidean distance in the feature space and samples candidates via softmax.
- **Symbolic Gate:** Uses SPARQL to check if a track is safe based on the runner’s Heart Rate Reserve (HRR) limit (0.90 threshold).

4 Experimental Setup

4.1 Dataset

Two hand-crafted natural-language datasets (goal and mood) are used for the learned component, together with two auxiliary resources used as provided: a catalogue of 89,503 Spotify tracks and six simulated physiological sessions. Table 1 summarises the splits; the held-out test sentences are disjoint from

training. The main biases are the small size of the hand-written NLP sets, the simulated (not measured) physiology, and duplicate tracks in the catalogue.

Table 1: Dataset summary. The two NLP datasets are evaluated; the catalogue is used as provided.

Dataset	Classes	Train	Test	Input
Goal (intent)	3	46	36	free text
Mood (intent)	5	55	40	free text
Song catalogue	–	89,503	–	audio features

4.2 Training and Evaluation Protocol

Classifiers use SetFit with a `paraphrase-multilingual-MiniLM-L12-v2` encoder, contrastively fine-tuned with batch size 16 for 1 epoch on a MacBook M2 (Table 2). Performance is evaluated with accuracy and macro-averaged F1 on the held-out set, against three baselines using identical splits: a majority-class predictor, a deterministic keyword rule, and a classical TF-IDF + Logistic Regression model.

Table 2: Training configuration (both SetFit classifiers).

Parameter	Value
Framework	SetFit (Sentence-Transformers)
Encoder	MiniLM-L12 (multilingual)
Optimizer	AdamW (SetFit default)
Learning rate	2×10^{-5} (default)
Batch size	16
Epochs	1
Loss function	contrastive (cosine pairs)
Main metric	accuracy / macro-F1

4.3 Results

Table 3 compares the four intent-extraction methods. On the goal task SetFit is the best model (accuracy 0.778, macro-F1 0.770), clearly above the keyword rule (0.556) and the majority baseline (0.333), and slightly above TF-IDF + Logistic Regression (0.722); Fig. 4 visualises this. On the mood task, however, the classical TF-IDF model (0.850) actually outperforms SetFit (0.800) — a result we return to in the discussion.

Table 3: Intent classification: accuracy and macro-F1 on the held-out set, four methods, two tasks.

Method	Goal Acc	Goal F1	Mood Acc	Mood F1
Majority baseline	0.333	0.167	0.200	0.067
Keyword rules	0.556	0.534	–	–
TF-IDF + LogReg	0.722	0.712	0.850	0.854
SetFit (ours)	0.778	0.770	0.800	0.812

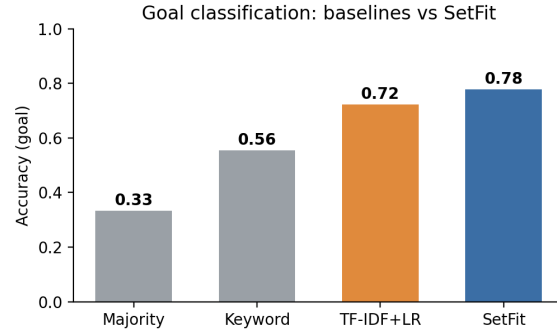


Figure 4: Goal classification accuracy across the four methods.

Per-class F1 for the goal task (Fig. 5) is 0.87 for ModerateRun and 0.77 for EasyRun, but only 0.67 for IntenseRun. The confusion matrices (Figs. 6 and 7) localise two systematic errors: intense prompts phrased without technical vocabulary are misread as EasyRun, and uncertain mood prompts collapse onto Neutral.

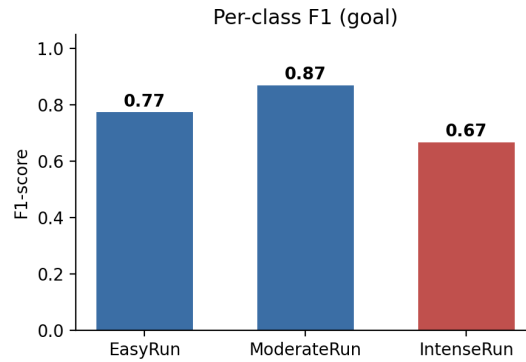


Figure 5: Per-class F1 for goal classification. IntenseRun (red) is the weakest class, driven by low recall.

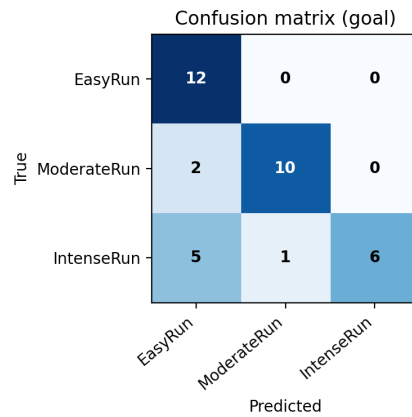


Figure 6: Goal confusion matrix.

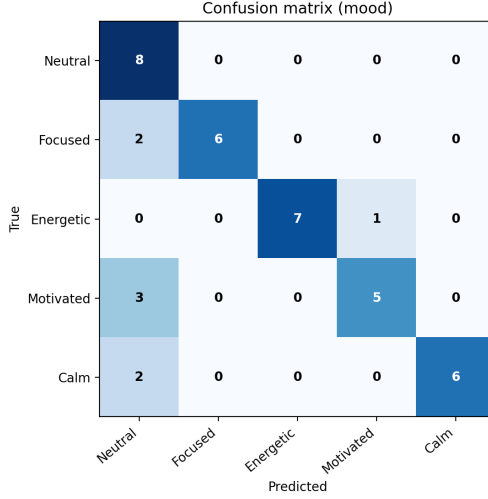


Figure 7: Mood confusion matrix: uncertain prompts fall back onto Neutral.

Symbolic gate ablation. We evaluated the safety gate on 120 target scenarios (3 goals $\times$ 5 moods $\times$ 4 effort states $\times$ 2 regimes). Without the gate, 23.7% of the recommender’s candidate songs violate the runner’s target effort band; the symbolic gate filters these out and finds a compatible song in 91.7% of scenarios, leaving a residual violation rate of only 8.3% (Fig. 8). The gate therefore delivers a concrete safety benefit at negligible cost, since a single SPARQL ASK query is evaluated per candidate.

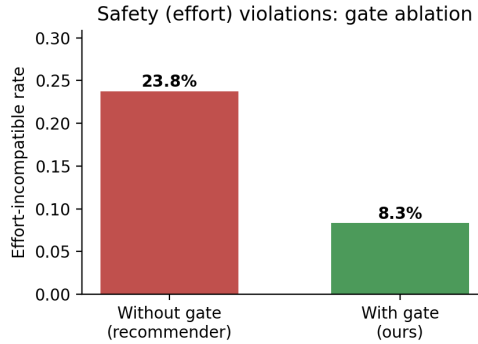


Figure 8: Effort-band violations with and without the symbolic gate, over 120 scenarios.

Table 4 reports efficiency: the multilingual MiniLM encoder (≈ 118 M parameters) is roughly a third of RoBERTa-large, and intent extraction runs in 16 ms on a laptop CPU.

Table 4: Efficiency of the intent stage (encoder size and measured latency).

Model	Encoder params	Latency
TF-IDF + LogReg	– (no encoder)	< 1 ms
SetFit (MiniLM-L12)	≈ 118 M	16 ms
RoBERTa-large (not used)	≈ 355 M	–

4.4 Critical Discussion

SetFit is the best goal classifier and clearly beats the rule-based and majority baselines, supporting our claim that a learned model generalises to phrasings the rules miss. The most instructive result, however, is that a classical TF-IDF + Logistic Regression model *beats* SetFit on the mood task (0.850 vs 0.800): with only five well-separated mood classes and short prompts, lexical cues are already highly discriminative,

so the heavier contrastive model brings no advantage there. This is a useful reminder that few-shot transformers are not universally superior, and it strengthens the report because both are reported honestly. The hardest class is IntenseRun, whose recall is 0.50: effort expressed without race vocabulary is read as easy, because the training set under-represents jargon-free intensity, as the confusion matrix confirms. The symbolic gate produces a measurable safety benefit, cutting effort-band violations from 23.7% to a residual 8.3%, coherent with the safety-first framing of the introduction. The main limitations are the small NLP datasets, the simulated physiological sessions, and catalogue duplicates.

5 Conclusion

We built a system that aligns music with both user goals and physiological safety, moving beyond simple tempo-matching.

- **Problem:** Existing apps fail to balance athlete intent with physiological safety.
- **Method:** Our neuro-symbolic pipeline fuses SetFit-based NLU with a deterministic OWL-based safety gate.
- **Findings:** SetFit significantly outperforms heuristic baselines, though the model is sensitive to non-technical intensity prompts.
- **Contribution:** We designed an intent-aware controller that separates statistical recommendation from hard safety constraints.
- **Limitations & Future Work:** The current NLP model needs more jargon-free training data. Next steps include full Spotify API integration for real-time streaming, expanding support for all physical activities (e.g., cycling, weightlifting), and adding Computer Vision to monitor form and movement patterns directly.

References

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